# Basics of Computer Grapics and Its Importance

By Dheeraj Karki

# Motivation

# Form and Formless

Life is a combination of form and formless. Feelings have no form but their expressions have form. The Self has no form but its abode has form. Similarly, wisdom and grace have no form but are expressed through form. Discarding the formless, you become inert, materialistic and paranoid. Discarding the form, you become a lost ascetic, a space cadet or an emotional junk!

Form and formless are intertwined in this world. The formless can only be expressed in form and form can only be thought with the formless.

Swami Vivekananda

PICTUREQUOTES.com

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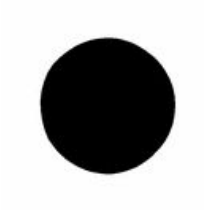
# Some Basic Concept

# What is Point?

A geometrical point is a location in space. It has no other characteristics. It has no length, width, or thickness. It is pure location.



# Point

- The above answer would not help you if you somehow did not already understand the concept of "point". The phrase "location in space" designates the same concept as the word "point". The definition cannot convey a concept you do not already have.
- In geometry, the word point is not defined. It is one of the undefined primitives that are used to define other objects. Instead of defining "point" books give examples and hope that you somehow build up the concept from them.

# Physical Point

- A point is an idealization. On the scale of a building, the end of a spike is small enough to be considered an exact location, a point. But on the scale of your desk, its end is too blunt to exactly define one location. At the scale of a desk, the spike would have to taper to a pinpoint. Then, perhaps, you could consider its end to define a point.
- But if you put that pin point under a microscope, then for that scale it is again too blunt. It would have to taper further, perhaps to the thickness of a bacterium, before the end could define a point. But now, on the scale of bacteria, the end of the spike is again too thick to exactly define a location.

# Physical Point

- So, points in 3D space are an idealization. For a computer model of a building, the sharp end of a spike is exact enough to be a point. The edges, planes, and other shapes that make up the building do not need to be specified to any greater accuracy.

# What is Line?

A line segment is the straight path between two points. It has no thickness. There is only one line segment between the two points.

# Line

- Again, if you did not already have the concept of "line" the above definition would not be enough to learn it. The phrase "straight path" designates the concept that is being defined. In 3D graphics you mostly deal with line segments. In geometry, lines continue without end. A line segment is the part of a line between two points.
- In geometry books the world line is usually not defined. It (like point) is an undefined concept used to define other objects. Books give pictures and examples of lines and hope that somehow you will come to understand what a line is by looking at them.
- Often people say "line" when properly they should say "line segment."

# Line

- At the scale of the building, a thin wire stretched between the two spikes is a close approximation to a line segment.
- Again, this is a matter of scale. The wire is likely to be thicker than a line drawn by a pencil. On your desk the wire is too thick to be considered a line segment. Perhaps a thin, straight pencil line is thin enough to be considered a geometrical line. If that is too thick, then a length of spider silk stretched between two points might work.
- But on an architectural level, taught wires can be considered line segments. You could build a geometrical model of the building using line segments to represent edges.

# Plane

Our world has three dimensions, but there are only two dimensions on a plane:

- length and width make a plane
- x and y also make a plane

A plane has no thickness, and goes on forever.

So, What you think what we need to draw a Plane?

# Triangle

A triangle is made by connecting three points in 3D space with line segments. Unless the three points all lie on the same line, three points in space define a unique triangle. Triangles in space are important in 3D graphics. The 3D objects that make up the scene are constructed out of triangles and other flat polygons.

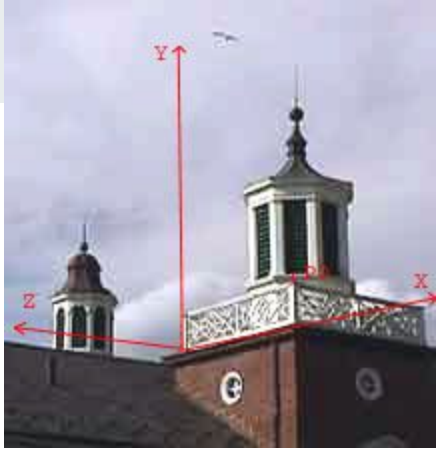
Why we are not able to define its Position? What we are missing here?

Can you let me the location of Marker on Table?

How can you define its Position?

# Coordinate System

# Representation




- You can think about the points, lines, and plane figures that compose the 3D scene without using a coordinate system. In fact, that is what we have been doing. But in computer graphics we need to represent these objects, somehow, and to manipulate them. To do this, we need a method to represent such objects.
- Pick a convenient point in the 3D scene and call it the origin. Now define three lines called X, Y, and Z that go through the origin. Usually the three lines are perpendicular to each other. Each line is called an axis.
- Pick a positive direction for each line. There are several choices here, but let's make the choice shown in the picture.
- Now each point of the X axis is a unique distance (positive, negative, or zero) from the origin.

# Transformation-2D

# Let Solve the Problem

- Let we have equation
- P = (1- t)^3  P1 + 3t(1-t)^2 P2 + 3t^2 P3+ t^3 P4
- If you want to solve this equation How will you solve. Let suppose you have all different t values.

# Great! I hope we are able to solve it.

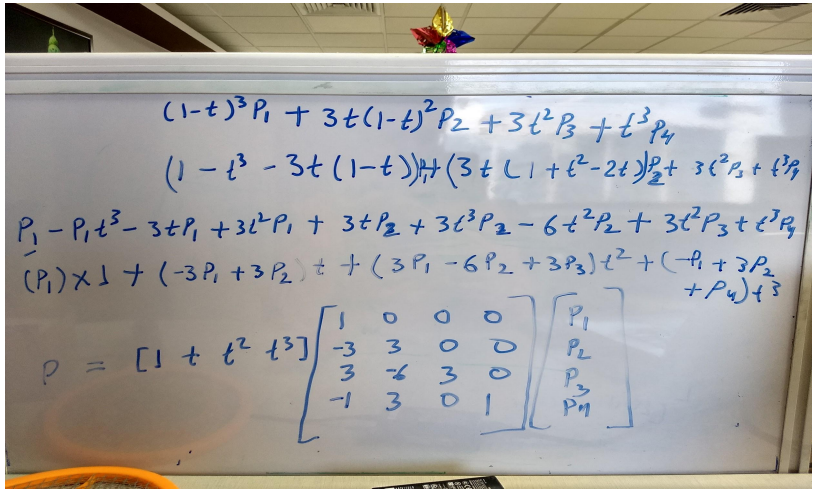
- Now think how computer will solve it?

No, Not a good Idea! What is good Solution?.

Will Traditional way to calculate above problem good for Computer?

Magic of Matrix!!

# After solving above equation we can get this result in Matrix.



# Linear Transformation

If T is linear transformation and u and v are the vectors and c is the scalar, then following hold

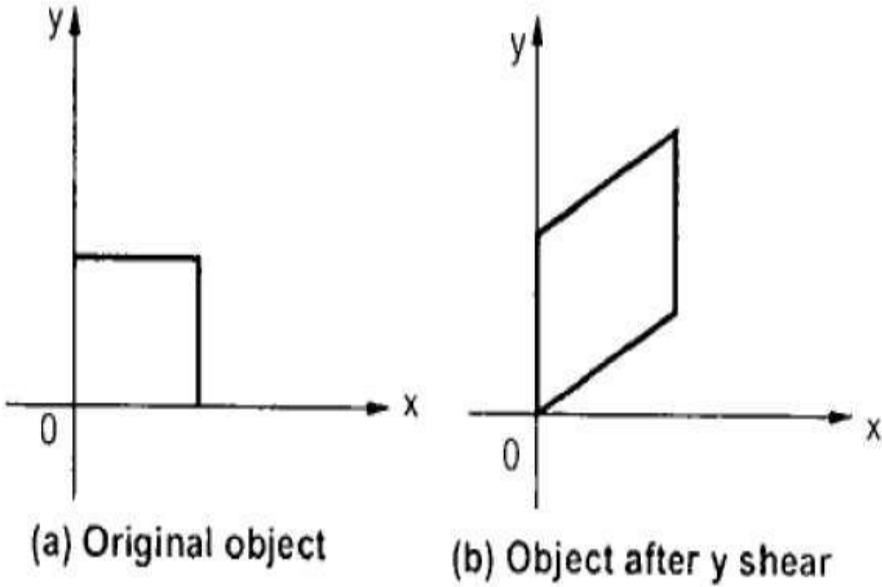
- $T(u+v) = T(u) + T(v)$
- $T(cv) = cT(v)$

Example: Rotation, Scaling

But what about translation?
Translation is not Linear Transformation.

# Affine Transformation

- Combine Linear Transformation with Translation
- Origin doesn't necessarily map to Origin
- Lines map to Lines
- Parallel lines all remain Parallel.
- Ratios are preserved.



# Vector Space

Vector space consist two important entities.

- <u>**Vector**</u>
  - a quantity having direction as well as magnitude, especially as determining the position of one point in space relative to another.
- <u>**Frame**</u>
  - Sometime they are also called coordinate system.
  - Consist subentities: 3 Vectors + Origin

# Affine Space

An affine space is a geometric structure that generalizes the properties of Euclidean spaces in such a way that these are independent of the concepts of distance and measure of angles, keeping only the properties related to parallelism and ratio of lengths for parallel line segments.

# Vector Space vs Affine Space

A vector space is an abstraction of how geometrical vectors (in the plane, say) behave. You can form linear combinations of vectors. Not all vectors look alike; they can vary in direction and in magnitude, and in particular there is the zero vector which is special.

An affine space is an abstraction of how geometrical points (in the plane, say) behave. All points look alike; there is no point which is special in any way. You can't add points. However, you can subtract points (giving a vector as the result). And you can form convex combinations of points.

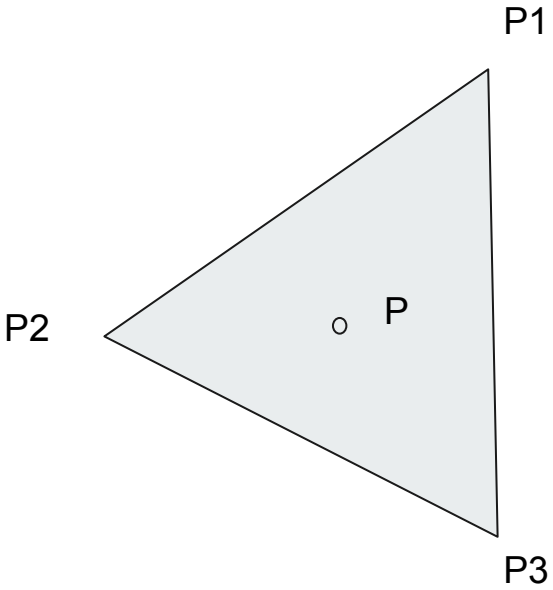
# Point and Vector

- Adding Point and Vector will give you Point : P + V = Q
- Subtracting two points will give you Vector: P - Q = V
- Generalized Vector and point representation: P + αV = Q
  - Such that you can find any point on vector v by changing α value.
- Affine Combination
  - P1 + t(P2 - P1) => (t-1)P1 + tP2
  - So, P = t1P1 + t2P2 + t3P3, such that t1+t2+t3 = 1
- Affine Space: Points + Vectors

# Confused till now

https://blogs.msdn.microsoft.com/rezanour/2011/06/26/math-primer-series-vectors-iii-affine-spaces-linear-and-affine-combinations/

# How to write point P in terms of vertices of triangle?



# How to write point P in terms of vertices of triangle?

- $P = P1 + t2(P2 - P1) + t3( P3 - P1)$
- $P = (1 - t2 - t3) P1 + t2P2 + t3P3$
- Let $1 - t2 - t3 = t1$
- So we have, $P = t1P1 + t2P2 + t3P3$
- Such that $t1 + t2 + t3 =1$ // Affine Combination
- We can think it as a coordinate system of triangle.
- Called Barycentric Coordinate

# Example

# Example

http://rawgit.com/Dheeraj-Karki/WebGLExperiment/master/index.html